INCH-POUND

MIL-PRF-19500/713
28 April 2005

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED
(TOTAL DOSE AND SINGLE EVENT EFFECTS)
TRANSISTORS, P-CHANNEL, SILICON, TYPES 2N7549T1, 2N7549U2, 2N7550T1, AND 2N7550U2,
JANTXVR, F AND JANSR, F

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of
this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for an P-channel, enhancement-mode, MOSFET, radiation hardened (total dose and single event effects (SEE)), power transistor. Two levels of product assurance are provided for each device type as specified in MIL-PRF-19500, with avalanche energy maximum rating (E_{AS}) and maximum avalanche current (I_{AS}).

1.2 Physical dimensions. See figure 1, TO-254AA (T1) and figure 2, SMD2 TO-276AC (U2).

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	P_T (1) T_C = +25°C	P_T T_A = +25°C (free air)	$R_{\theta JC}$	V_{DS}	V_{DG}	V_{GS}	I_{D1} (2) (3) T_C = +25°C	I_{D2} (2) (3) T_C = +100°C	I_S	I_{DM} (4)	T_J and T_{STG}	V_{ISO} 70,000 foot altitude
	W	W	°C/W	V dc	V dc	V dc	A dc	A dc	A dc	A(pk)	°C	V dc
2N7550T1	208	2.6	0.60	-100	-100	±20	-45	-28.5	-45	-180	-55	100
2N7550U2	250	1.6	0.50	-100	-100	±20	-47	-30	-47	-188	to	100
2N7549T1	208	2.6	0.60	-200	-200	±20	-30	-19	-30	-120	+150	200
2N7549U2	250	1.6	0.50	-200	-200	±20	-33.5	-21	-33.5	-134		200

(1) Derate linearly by 2.0 W/°C (U2) or 1.67 W/°C (T1) for T_C > +25°C.
(2) The following formula derives the maximum theoretical I_D limit. I_D is limited by package design and device construction, to 45A for T1 or to 56A for U2:

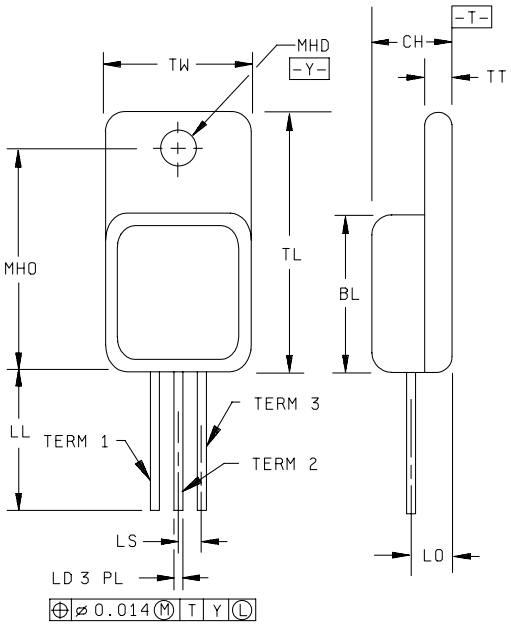
$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

(3) See figure 3, maximum drain current graph.
(4) I_{DM} = 4 X I_{D1}, as defined in note (2).

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

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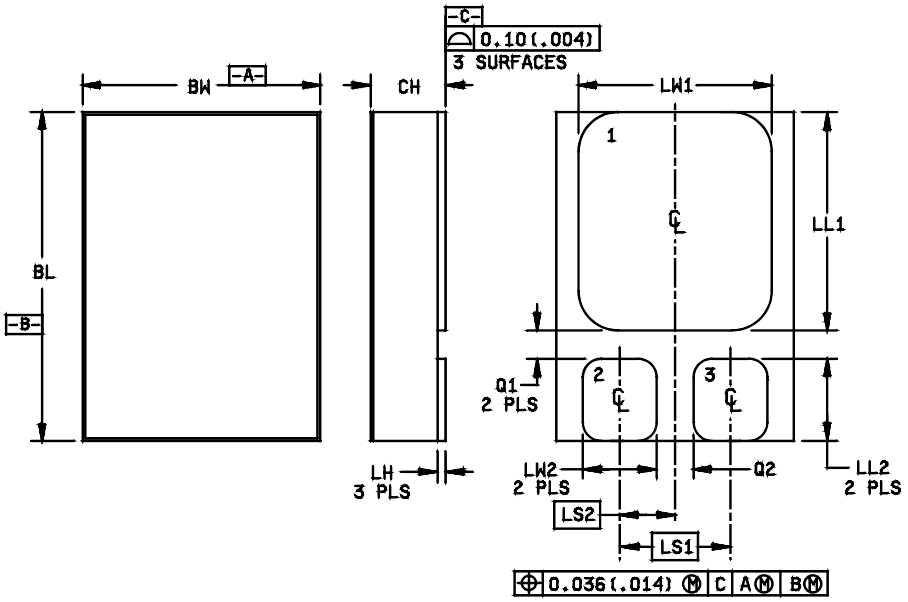


Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BL	.535	.545	13.59	13.84	
CH	.249	.260	6.32	6.60	
LD	.035	.045	0.89	1.14	
LL	.510	.570	12.95	14.48	3
LO	.150 BSC		3.81 BSC		
LS	.150 BSC		3.81 BSC		
MHD	.139	.149	3.53	3.78	
MHO	.665	.685	16.89	17.40	
TL	.790	.800	20.07	20.32	4
TT	.040	.050	1.02	1.27	
TW	.535	.545	13.59	13.84	4
Term 1	Drain				
Term 2	Source				
Term 3	Gate				

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Protrusion thickness of ceramic eyelets included in dimension LL.
4. All terminals are isolated from case.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for TO-254AA (2N7549T1 and 2N7550T1).

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Min	Min	Max
BL	.395	.405	10.04	10.28
BW	.291	.301	7.40	7.64
CH	.1085	.1205	2.76	3.06
LH	.010	.020	0.25	0.51
LW1	.281	.291	7.14	7.41
LW2	.090	.100	2.29	2.54
LL1	.220	.230	5.59	5.84
LL2	.115	.125	2.93	3.17
LS1	.150 BSC		3.81 BSC	
LS2	.075 BSC		1.91 BSC	
Q1	.030		0.762	
Q2	.030		0.762	
TERM 1	Drain			
TERM 2	Gate			
TERM 3	Source			

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The lid shall be electrically isolated from the drain, gate and source.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions for SMD2 TO-276AC (2N7550U2 and 2N7549U2).

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1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(TH)}$ $V_{DS} \geq V_{GS}$ I_D = 1.0 mA dc		Max I_{DSS1} V_{GS} = 0 V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(ON)}$ (1) V_{GS} = 12 V dc		E_{AS} at I_{D1}	I_{AS}
					T_J = +25°C at I_{D2}	T_J = +150°C at I_{D2}		
	V dc	V dc		µA dc	ohm	ohm	mJ	A
		Min	Max					
2N7550T1	-100	-2.0	-4.0	-10	0.050	0.103	480	-45
2N7550U2	-100	-2.0	-4.0	-10	0.049	0.113	400	-47
2N7549T1	-200	-2.0	-4.0	-10	0.103	0.232	332	-30
2N7549U2	-200	-2.0	-4.0	-10	0.102	0.221	303	-33.5

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 (T1, TO-254AA) and 2 (U2, surface mount TO-276AC) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted to meet the requirements of this specification.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection.

3.5.1 Handling. MOS devices shall be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate shall be terminated to source, $R \leq 100\ k\Omega$, whenever bias voltage is to be applied drain to source.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking may be omitted from the body, but shall be retained on the initial container.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 SEE. Design capability shall be tested on the initial qualification and thereafter whenever a major die design or process change is introduced. End-point measurements shall be in accordance with table III.

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4.3 Screening (JANS and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500) (1) (2)	Measurement	
	JANS level	JANTXV level
(3)	Gate stress test (see 4.3.1)	Gate stress test (see 4.3.1)
(3)	Method 3470 of MIL-STD-750, E_{AS} (see 4.3.2)	Method 3470 of MIL-STD-750, E_{AS} (see 4.3.2)
(4) 3c	Method 3161 of MIL-STD-750, thermal impedance (see 4.3.3)	Method 3161 of MIL-STD-750, thermal impedance (see 4.3.3)
7	Optional.	Optional.
9	Subgroup 2 of table I herein; I_{GSSF1}, I_{GSSR1}, I_{DSS1}	Not applicable
10	Method 1042 of MIL-STD-750, test condition B	Method 1042 of MIL-STD-750, test condition B
11	Subgroup 2 of table I herein; I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(TH)1}$ ΔI_{GSSF1} = ± 20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 10 μA dc or ± 100 percent of initial value, whichever is greater.	Subgroup 2 of table I herein; I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(TH)1}$
12	Method 1042 of MIL-STD-750, test condition A	Method 1042 of MIL-STD-750, test condition A
13	Subgroups 2 and 3 of table I herein; ΔI_{GSSF1} = ± 20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 10 μA dc or ± 100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20 percent of initial value. $\Delta V_{GS(TH)1}$ = ± 20 percent of initial value.	Subgroups 2 and 3 of table I herein; ΔI_{GSSF1} = ± 20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 10 μA dc or ± 100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20 percent of initial value. $\Delta V_{GS(TH)1}$ = ± 20 percent of initial value.
14	Required.	Required.

(1) At the end of the test program, I_{GSSF1}, I_{GSSR1}, and I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1} and $V_{GS(TH)1}$ shall be invoked.
(3) Shall be performed anytime before screen 9.
(4) This test shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements

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4.3.1 Gate stress test. Apply V_{GS} = 30 V, minimum for t = 250 μs, minimum.

4.3.2 Single pulse avalanche energy (E_{AS}).

a. Peak current ..... $I_{AS} = I_{D1}$.

b. Inductance ..... $L = \left[\frac{2E_{AS}}{(I_{D1})^2}\right]\left[\frac{V_{BR} - V_{DD}}{V_{BR}}\right]$ mH minimum.

c. Gate to source resistor R_{GS} ..... $25\ \Omega \leq R_{GS} \leq 200\ \Omega$.

d. Supply voltage ..... V_{DD} = 50 V dc

e. Initial case temperature ..... T_C = +25° C, -5° C, +10° C.

f. Gate voltage ..... V_{GS} = 12 V dc.

g. Number of pulses to be applied ..... 1 pulse minimum.

4.3.3 Thermal impedance. The thermal impedance measurement shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit (not to exceed figure 4, thermal impedance curves and the table I, subgroup 2 limits) for thermal impedance in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot, to be plotted on the applicable X bar R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for engineering evaluation and disposition. This procedure may be used in lieu of an in line process monitor.

a. Measuring current (I_M) ..... 10 mA.

b. Drain heating current (I_H) ..... 11.11 A (T1) or 13.88 A (U2)

c. Heating time (t_H) ..... 100 ms (T1) or 20 ms (U2)

d. Drain-source heating voltage (V_H) ..... 12 V.

e. Measurement time delay (t_{MD}) ..... 30 - 60 μs.

f. Sample window time (t_{SW}) ..... 10 μs maximum.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500 and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1051	Test condition G, 100 cycles.
B3	2077	SEM (scanning electron microscope).
B4	1042	Intermittent operation life, condition D, 2,000 cycles. No heat sink or forced-air cooling on the device shall be permitted during the on cycle. t_{on} = 30 seconds minimum.
B5	1042	Accelerated steady-state gate bias, condition B, V_{GS} = rated; T_A = +175°C, t = 24 hours minimum; or T_A = +150°C, t = 48 hours minimum.
B5	1042	Accelerated steady-state reverse bias, condition A, V_{DS} = rated; T_A = +175°C, t = 120 hours minimum; or T_A = +150°C, t = 240 hours minimum.
B5	2037	Bond strength, test condition A.

4.4.2.2 Group B inspection, table VIb (JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B2	1051	Test condition G, 25 cycles.
B3	1042	Intermittent operation life, condition D, 2,000 cycles. No heat sink or forced-air cooling on the device shall be permitted during the on cycle. t_{on} = 30 seconds minimum.
B5 and B6		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Test condition A; weight = 10 pounds; t = 15 s.
C5	3161	See 4.5.2.
C6	1042	Intermittent operation life, condition D, 6,000 cycles. No heat sink or forced-air cooling on the device shall be permitted during the on cycle. t_{on} = 30 seconds minimum.

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table VIII of MIL-PRF-19500 and table II herein.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\theta JC(max)}$ = 0.60 °C/W (T1) or 0.50 ºC/W (U2). The following parameter measurements shall apply:

a. Measuring current (I_M) ..... 10 mA.

b. Drain heating current (I_H) ..... 11.11 A (T1) or 13.88 A (U2).

c. Heating time (t_H) ..... Steady-state (see method 3161 of MIL-STD-750, for definition).

d. Drain-source heating voltage (V_H) ..... 12 V.

e. Measurement time delay (t_{MD}) ..... 30 to 60 μs.

f. Sample window time (t_{SW}) ..... 10 μs maximum.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.3.3	$Z_{\theta JC}$			
2N7550T1, 2N7549T1					0.45	°C/W
2N7550U2, 2N7549U2					0.40	°C/W
Breakdown voltage, drain to source	3407	V_{GS} = 0 V dc, I_D = -1 mA dc, bias condition C	$V_{(BR)DSS}$			
2N7550T1 and U2				-100		V dc
2N7549T1 and U2				-200		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = -1 mA dc	$V_{GS(th)1}$	-2.0	-4.0	V dc
Gate reverse current	3411	V_{GS} = +20 V dc, bias condition C, V_{DS} = 0	I_{GSSF1}		+100	nA dc
Gate reverse current	3411	V_{GS} = -20 V dc, bias condition C, V_{DS} = 0	I_{GSSR1}		-100	nA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		-10	µA dc
Static drain to source on-state resistance	3421	V_{GS} = -12 V dc, condition A, pulsed (see 4.5.1), I_D = I_{D2}	$r_{DS(on)1}$			
2N7550T1					0.050	Ω
2N7550U2					0.049	Ω
2N7549T1					0.103	Ω
2N7549U2					0.102	Ω
Forward voltage	4011	Pulsed (see 4.5.1), I_D = I_{D1}, V_{GS} = 0 V dc	V_{SD}		-5.0	V
Subgroup 3						
High-temperature operation:		T_C = T_J = +125°C				
Gate reverse current	3411	V_{GS} = -20 V dc and +20 V dc, bias condition C, V_{DS} = 0	I_{GSS2}		± 200	nA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS2}		-25	µA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - continued						
High-temperature operation:		$T_C = T_J = +125°C$				
Static drain to source on-state resistance	3421	V_{GS} = -12 V dc, pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(on)3}$			
2N7550T1					0.098	Ω
2N7550U2					0.098	Ω
2N7549T1					0.230	Ω
2N7549U2					0.204	Ω
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = -1 mA dc	$V_{GS(th)2}$	-1.0		V dc
Low-temperature operation:		$T_C = T_J = -55°C$				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = -1 mA dc	$V_{GS(th)3}$		-5.0	V dc
Subgroup 4						
Forward transconductance	3475	I_D = rated I_{D2}, V_{DD} = -15 V (see 4.5.1)	g_{FS}			
2N7550T1 and U2				24		S
2N7549T1 and U2				23		S
Switching time test	3472	I_D = rated I_{D1}, V_{GS}= -12 V dc, R_G = 1.2 Ω (T1) or 2.35 Ω (U2), V_{DD} = 50 percent of rated V_{DS}				
Turn-on delay time			$t_{d(on)}$			
2N7550T1					35	ns
2N7550U2					30	ns
2N7549T1					35	ns
2N7549U2					35	ns
Rise time			t_r			
2N7550T1					140	ns
2N7550U2					100	ns
2N7549T1					50	ns
2N7549U2					80	ns
Turn-off delay time			$t_{d(off)}$			
2N7550T1					70	ns
2N7550U2					70	ns
2N7549T1					75	ns
2N7549U2					100	ns

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Fall time			t_f			
2N7550T1					45	ns
2N7550U2					170	ns
2N7549T1					100	ns
2N7549U2					200	ns
Subgroup 5						
Safe operating area test (high voltage)	3474	See figures 5 and 6; t_p = 10 ms, V_{DS} = 80 percent of rated V_{DS}				
Electrical measurements		See table I, subgroup 2 herein.				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B	$Q_{G(on)}$			
2N7550T1 and U2					170	nC
2N7549T1					175	nC
2N7549U2					180	nC
On-state gate charge			Q_{GS}			
2N7550T1 and U2					65	nC
2N7549T1					75	nC
2N7549U2					60	nC
Gate to drain charge			Q_{GD}			
2N7550T1 and U2					30	nC
2N7549T1					70	nC
2N7549U2					40	nC
Reverse recovery time		di/dt ≤ 100A/μs, V_{DD} ≤ 50 V, I_D = I_{D1}	t_{rr}			
2N7550T1					200	ns
2N7550U2					230	ns
2N7549T1					300	ns
2N7549U2					450	ns

1/ For sampling plan, see MIL-PRF-19500.
2/ This test is required for the following end-point measurement only:
JANS, table VIa of MIL-PRF-19500, group B, subgroups 3 and 4; JANTXV, table VIb of MIL-PRF-19500, group B, subgroups 2 and 3; and table VII of MIL-PRF-19500, group C, subgroups 2 and 6, and table IX of MIL-PRF-19500, group E, subgroup 1.

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TABLE II. Group D inspection.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Pre-Irradiation Limits		Post-Irradiation Limits		Post-Irradiation Limits		Unit
	Method	Conditions		R, F		R		F		
				Min	Max	Min	Max	Min	Max	
Subgroup 1										
Not applicable										
Subgroup 2		T_C = +25°C								
Steady-state total dose irradiation (V_{GS} bias) 4/	1019	V_{GS} = -12V V_{DS} = 0								
Steady-state total dose irradiation (V_{DS} bias) 4/	1019	V_{GS} = 0 V_{DS} = 80 percent of rated V_{DS} (pre-irradiation)								
End-point electricals:										
Breakdown voltage, drain to source	3407	V_{GS} = 0 I_D = -1 mA bias cond. C	$V_{(BR)DSS}$							
2N7550T1, U2				-100		-100		-100		V dc
2N7549T1, U2				-200		-200		-200		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$	V_{GSth1}	-2.0	-4.0	-2.0	-4.0	-2.0	-5.0	V dc
Gate reverse current	3411	V_{GS} = -20 V V_{DS} = 0 bias cond. C	I_{GSSR1}		-100		-100		-100	nA dc
Gate forward current	3411	V_{GS} = 20 V V_{DS} = 0 bias cond. C	I_{GSSF1}		100		100		100	nA dc
Drain current	3413	V_{GS} = 0 bias cond. C V_{DS} = 80 percent of rated V_{DS} (pre-irradiation)	I_{DSS1}		-10		-10		-10	µA dc

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Pre-Irradiation Limits		Post-Irradiation Limits		Post-Irradiation Limits		Unit
	Method	Conditions		R, F		R		F		
				Min	Max	Min	Max	Min	Max	
Subgroup 2 - End-point electricals - Continued:										
Static drain to source on-state voltage	3405	V_{GS} = -12 V cond. A pulsed (see 4.5.1) $I_D = I_{D2}$	V_{DSon1}							
2N7550T1					1.425		1.425		1.425	V dc
2N7550U2					1.500		1.500		1.500	V dc
2N7549T1					1.975		1.975		1.975	V dc
2N7549U2					2.163		2.163		2.163	V dc
Forward voltage source to drain diode	4011	V_{GS} = 0 $I_D = I_{D1}$	V_{SD}		-5.0		-5.0		-5.0	V dc

1/ For sampling plan, see MIL-PRF-19500.
2/ Group D qualification may be performed anytime prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other specification sheet utilizing the same die design.
3/ At the manufacturer's option, group D samples need not be subjected to the screening tests, and may be assembled in it's qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.
4/ Separate samples shall be pulled for each bias.

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TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification inspection
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycling	1051	Test condition G, 500 cycles	
Hermetic seal Fine leak Gross leak	1071	 Test conditions G or H Test conditions C or D	
Electrical measurements		See table I, subgroup 2	
Subgroup 2 1/			12 devices c = 0
Steady-state gate bias	1042	Test condition B; 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Steady-state reverse bias	1042	Test condition A; 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Subgroup 4			Sample size N/A
Thermal impedance curves		Each supplier shall submit their qual-lot average and design maximum thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and thermal impedance limit shall be provided to the qualifying activity in the qualification report.	
Subgroup 5			
Not applicable			
Subgroup 6			3 devices c = 0
ESD	1020	Not required for devices classified as ESD class 1.	
Subgroup 8			22 devices c = 0
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476	Test conditions shall be derived by the manufacturer	

See footnotes at end of table.

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TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection
	Method	Conditions	
Subgroup 9			3 devices c = 0
SEE 2/ 3/ 4/	1080	See figure 7	
Electrical measurements 5/		I_{GSSF1}, I_{GSSR1}, and I_{DSS1} in accordance with table I, subgroup 2	
SEE irradiation		Fluence = 3E5 ±20 percent ions/cm^2 Flux = 2E3 to 2E4 ions/cm^2/sec Temperature = 25 ±5 °C	
		LET = 37.3 - 37.9 MeV-cm^2/mg Range = 33.1 - 36.8 microns Energy = 252.6 - 285 MeV	
2N7550T1, 2N7550U2		Insitu bias conditions: V_{DS} = -100 V and V_{GS} = 20 V	
2N7549T1, 2N7549U2		Insitu bias conditions: V_{DS} = -200 V and V_{GS} = 15 V V_{DS} = -75 V and V_{GS} = 20 V	
		LET = 59.7 - 59.9 MeV-cm^2/mg Range = 30.5 - 32.7 microns Energy = 314 - 345 MeV	
2N7550T1, 2N7550U2		Insitu bias conditions: V_{DS} = -100 V and V_{GS} = 15 V V_{DS} = -75 V and V_{GS} = 17.5 V V_{DS} = -25 V and V_{GS} = 20 V	
2N7549T1, 2N7549U2		Insitu bias conditions: V_{DS} = -200 V and V_{GS} = 10 V V_{DS} = -50 V and V_{GS} = 15 V	
		LET = 82.3 MeV-cm^2/mg Range = 28.4 - 28.5 microns Energy = 350 - 357 MeV	
2N7550T1, 2N7550U2		Insitu bias conditions: V_{DS} = -100 V and V_{GS} = 10 V V_{DS} = -30 V and V_{GS} = 15 V	
2N7549T1, 2N7549U2		Insitu bias conditions: V_{DS} = -200 V and V_{GS} = 10 V V_{DS} = -35 V and V_{GS} = 15 V	
Electrical measurements 5/		I_{GSSF1}, I_{GSSR1}, and I_{DSS1} in accordance with table I, subgroup 2	

1/ A separate sample for each test shall be pulled.
2/ Group E qualification of testing may be performed prior to lot formation. Qualification may be extended to other specification sheets utilizing the same structurally identical die design.
3/ Device qualification to a higher level LET is sufficient to qualify all lower level LET's.
4/ The sampling plan applies to each bias condition.
5/ Examine I_{GSS1} and I_{DSS1} before and following SEE irradiation to determine acceptability for each bias condition. Other test conditions in accordance with table I, subgroup 2, may be performed at the manufacturer's option.

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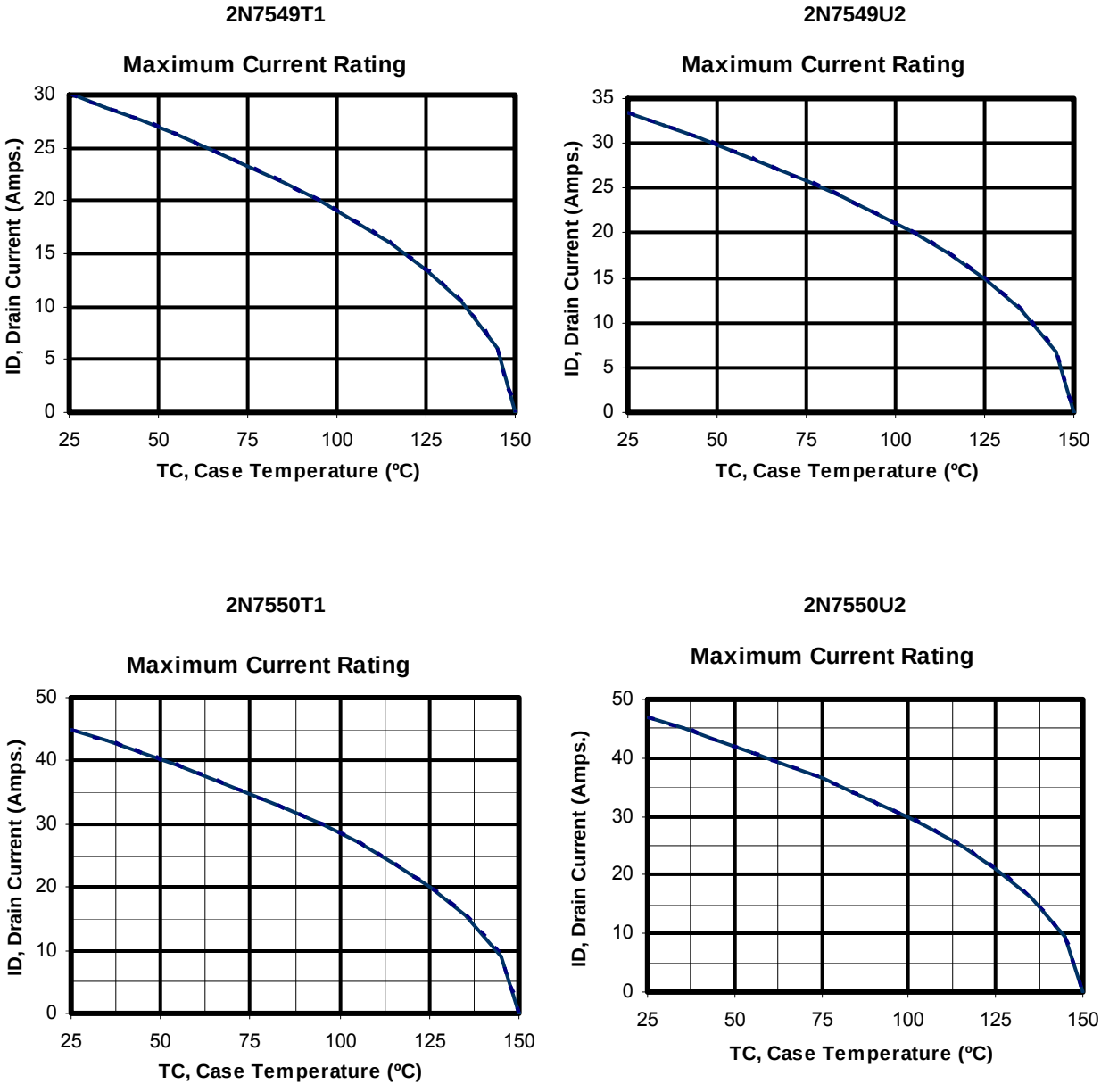
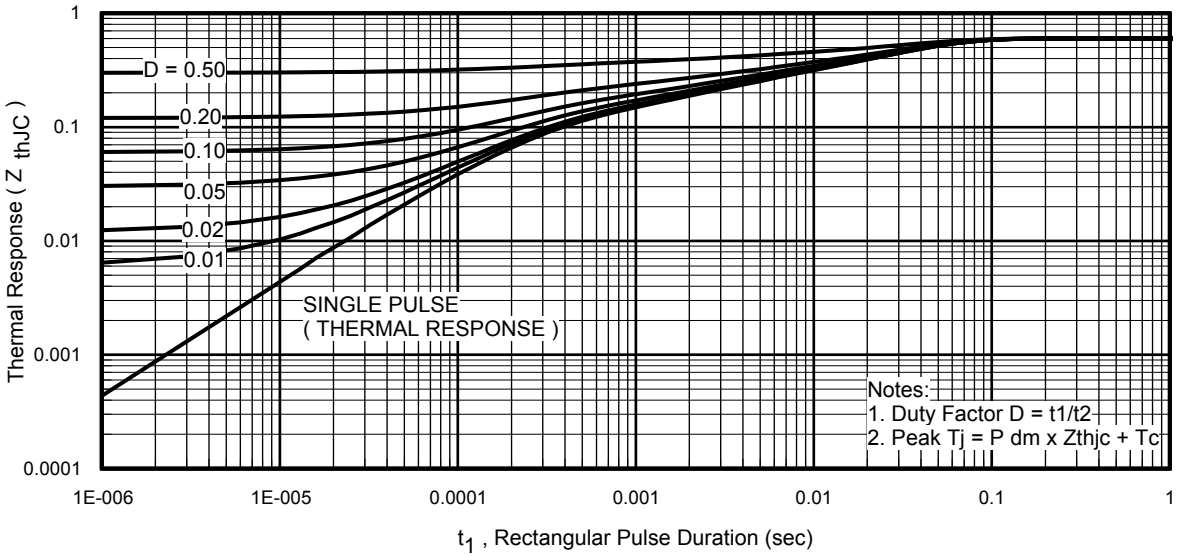


FIGURE 3. Maximum drain current vs case temperature graphs.

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2N7550T1 & 2N7549T1



2N7550U2 & 2N7549U2

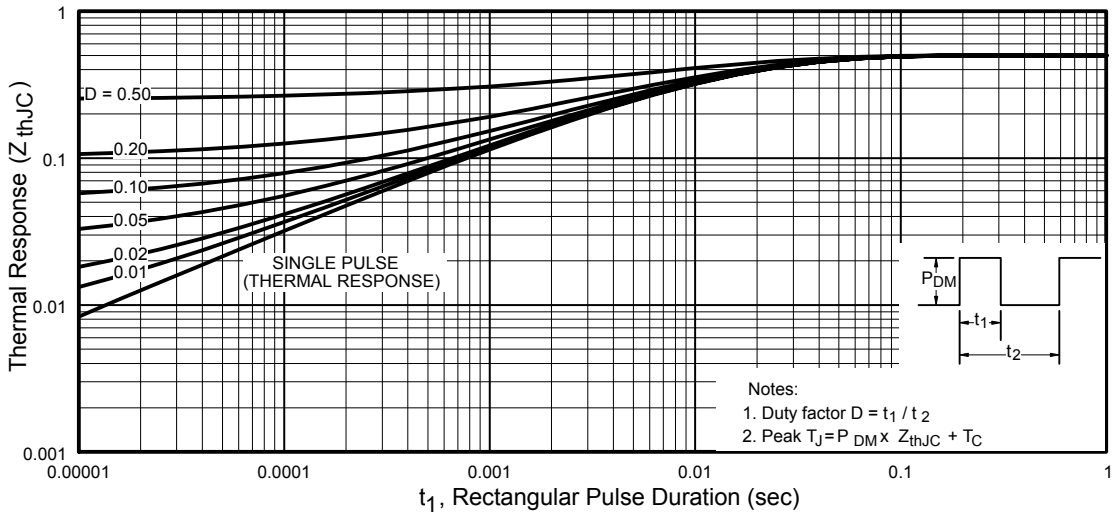
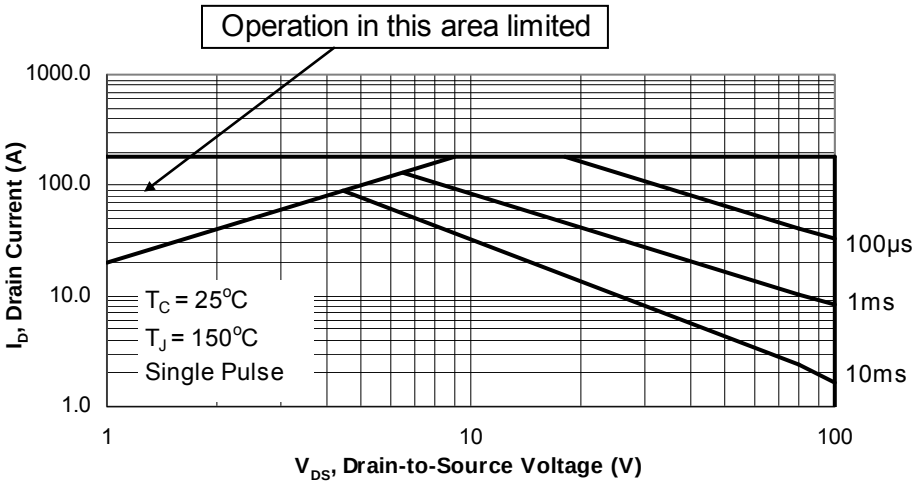


FIGURE 4. Thermal impedance curves.

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2N7550T1



2N7550U2

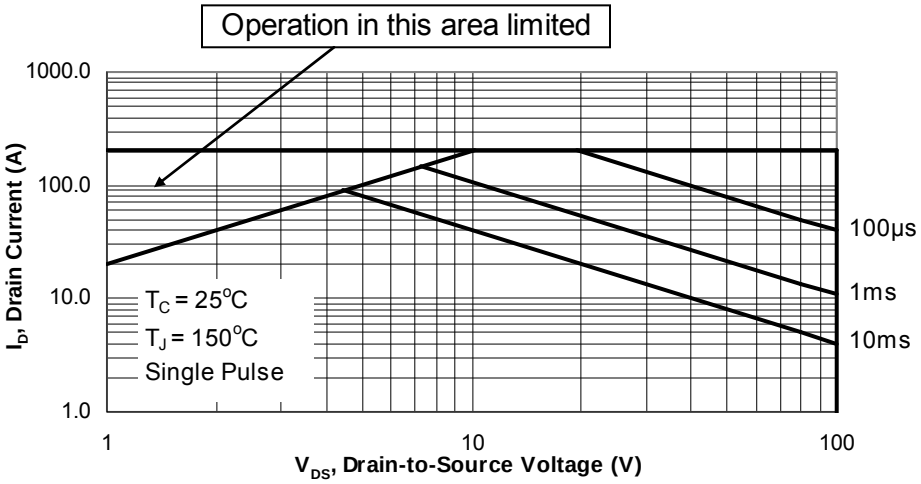
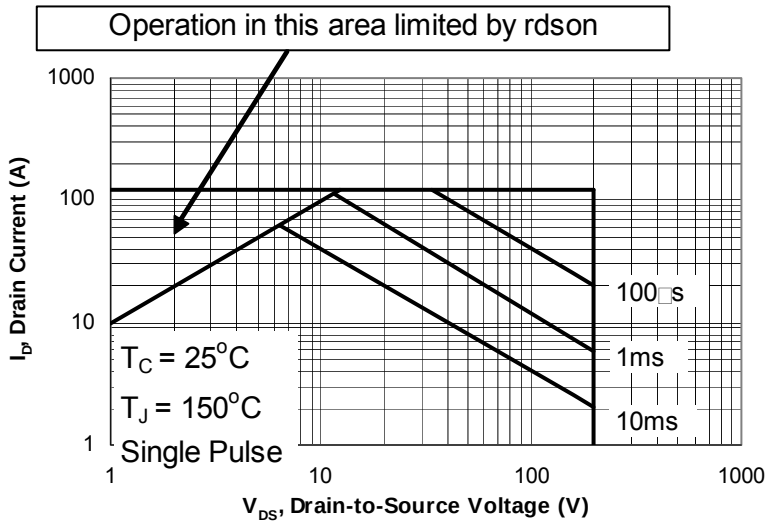


FIGURE 5. Safe operating area graph.

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2N7549T1



2N7549U2

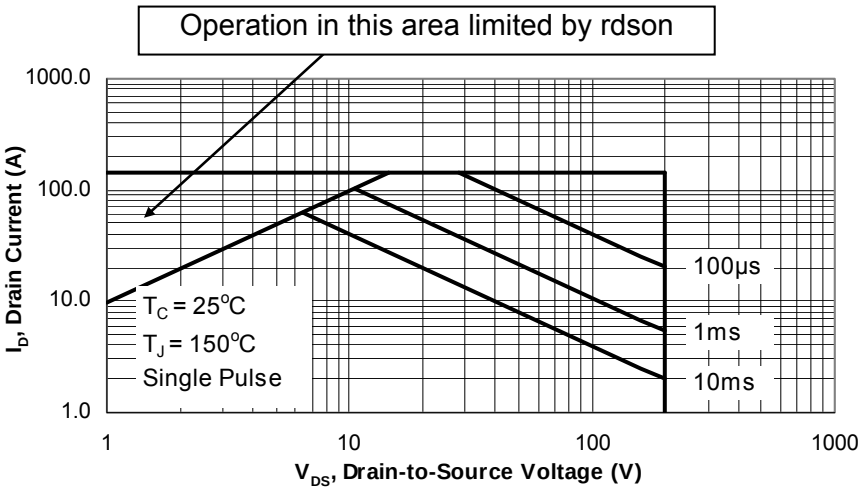
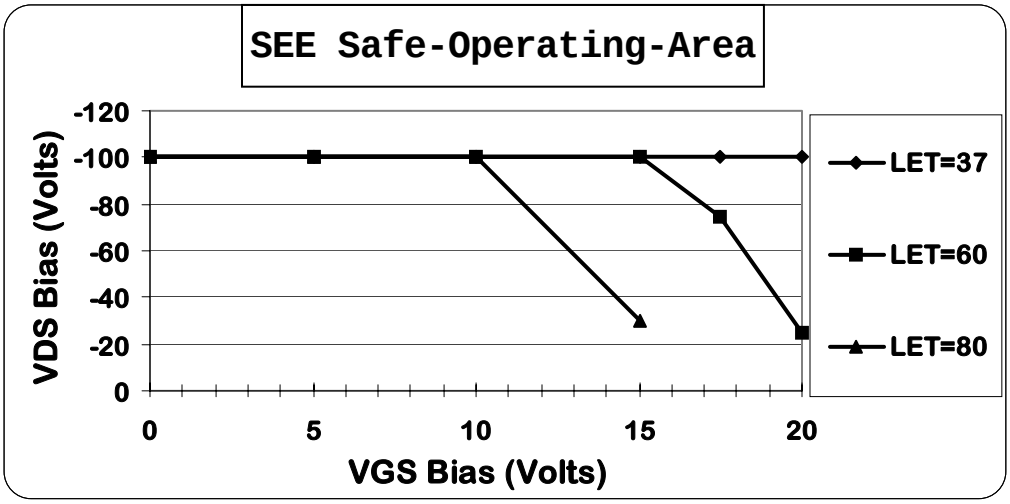


FIGURE 6. Safe operating area graph.

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2N7550T1 & 2N7550U2



2N7549T1 & 2N7549U2

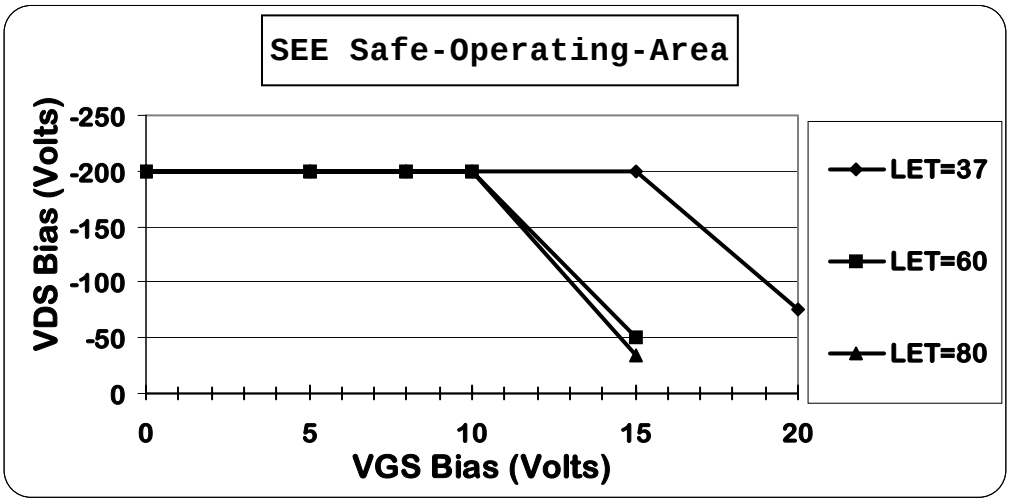


FIGURE 7. SEE safe operation area graph.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturer's PIN's are suitable for the military PIN.

Preferred types (military PIN)	Commercial PIN	
	TO-254AA	TO-276AC (SMD2)
2N7550T1	IRHMS59_160	
2N7550U2		IRHNA59_160
2N7549T1	IRHMS59_260	
2N7549U2		IRHNA59_260

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Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2995)

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